

LightingNet: An Integrated Learning Method for Low-Light Image Enhancement

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Abstract—Images captured in low-light environments suffer from serious degradation due to insufficient light, leading to the performance decline of industrial and civilian devices. To address the problems of noise, chromatic aberration, and detail distortion for enhancing low-light images using existing enhancement methods, this paper proposes an integrated learning approach (LightingNet) for low-light image enhancement. The LightingNet consists of two core components: 1) the complementary learning sub-network and 2) the vision transformer (ViT) low-light enhancement sub-network. ViT low-light enhancement sub-network is designed to learn and fit the current data to provide local high-level features through a full-scale architecture, and the complementary learning sub-network is utilized to provide global fine-tuned features through learning transfer. Extensive experiments confirm the effectiveness of the proposed LightingNet.

Index Terms—Generative adversarial network, low-light enhancement, vision transformer, learning transfer.

I. INTRODUCTION

WHEN performing a shooting task, most devices need to set the exposure time and sensitivity ISO (International Organization for Standardization). For shooting tasks in low-light environments, it is unavoidable to extend the exposure time or increase the ISO, as the illumination conditions of the environment itself are difficult to meet the shooting needs. The vast majority of current civilian equipment does not have the capacity for long-exposure shooting. In the case of extended exposure time, even the smallest amount of shake causes motion blur in the image, and sensitivity ISO often produces image noise that is difficult for the sensor to handle, resulting in severe loss of detail and image degradation. Advanced computer tasks that

rely on a variety of optical devices for detection and tracking can also degrade actual performance due to these degraded images captured in the dim environment [1], [72]. Low-light enhancement has a wide range of promising applications, and reliable low-light enhancement methods are important for improving the performance degradation due to insufficient light in a variety of sensors and shooting devices. Due to the performance of their optical devices and insufficient memory, it is very difficult for civilian devices to process images in dim environments. Consequently, low-light enhancement technologies have a huge research and development space in civilian mobile devices.

To solve the problem of serious degradation of images taken in low-light environments, early methods mainly focused on histogram equalization (HE). Although some methods based on histogram equalization managed to achieve a certain degree of enhancement of low-light images [2], [3], [4], low-light enhancement methods based on histogram equalization are limited by their theoretical limitations. These methods ignore image details while improving image contrast, often resulting in blurry images.

Retinex theory-based low-light enhancement methods have gained more attention compared to histogram equalization-based methods [7], [8], [9]. Retinex theory-based low-light enhancement methods use a prior or specific regularization to decompose the low-light image into two parts: illumination and reflection, and use the estimated reflection part as the enhancement result. However, Retinex theory-based low-light enhancement methods often struggle to find high-performance reliable priors and neglect the treatment of noise in the enhancement process, resulting in the loss of image details.

Along with the great success of convolutional neural networks (CNN) in various computer vision tasks, deep learning-based methods have made great progress in accuracy, speed, and enhancement effect in low-light enhancement tasks compared with traditional low-light enhancement methods [14], [15], [16]. Retinex theory-based deep learning methods have been further proposed, and several Retinex-based deep learning methods have shown stronger performance in low-light enhancement tasks [22], [31]. However, most deep learning methods still fail to balance noise control and brightening accuracy in the enhancement process, and the enhanced images are still accompanied by varying levels of artifacts, smearing, and detail loss. In this paper, to address the aforementioned problems, we propose a novel low-light enhancement method based on an integrated learning strategy. Specifically, we first construct the

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VIT low-light enhancement subnetwork with two key features: 1) a vision transformer (VIT) [38] network with a full-scale structure and 2) an improved self-attention PDTA (paired deconv transposed self-attention) block based on transposed attention [40]. Where PDTA generates transposed attention matrix by computing cross-channel covariance with additional depth-wise convolution in the tail to better decode the context, and fully connected layers are replaced by 3×3 convolution layer to offer a more refined pipeline. The full-scale architecture ensures that the feature information from all scales is available at each decoding layer of the network, and the mechanism enables the VIT low-light enhancement subnetwork to provide the local high-level information of the current training data. We then design a complementary learning subnetwork that provides globally robust complementary information to the overall network via the encoder structure of Res2Net pre-trained on ImageNet [49] and a specially constructed decoder structure. Our main contributions are as follows:

- An integrated learning-based generative adversarial network (LightingNet) for low-light enhancement is proposed, which consists of a VIT (vision transformer) low-light enhancement sub-network and a complementary learning sub-network based on learning transfer. The VIT low-light enhancement sub-network provides fine-grained high-level information on the current data, and the complementary learning sub-network provides global information with robust features.
- A VIT low-light enhancement sub-network with full-scale architecture in the low-light enhancement task is constructed, and an improved PDTA self-attention module for the VIT low-light enhancement subnetwork is also designed.
- We validate the feasibility of combining learning transfer based on traditional convolutional networks with VIT on low-light enhancement tasks. A complementary learning sub-network is constructed to combine the encoder section of Res2Net and a specially designed decoder section.
- Extensive experiments confirm the effectiveness of the proposed LightingNet over a large number of comparison methods in subjective and objective evaluation.

II. RELATED WORKS

In this section, the low-light enhancement methods can be categorized into three main categories: the histogram equalization-based methods, the Retinex-based methods, and the learning-based methods.

A. The Histogram Equalization-Based Methods

Histogram equalization-based methods, which transform the low-light image from a relatively concentrated gray-scale range to a uniform distribution over the entire range, were popular in the early stages of low-light enhancement tasks. Ibrahim et al. [2] proposed an extended method based on histogram equalization called brightness preserving dynamic histogram equalization (BPDHE), which could maintain the original brightness in the enhanced image. Celik et al. [3] designed a method that relied on the interpixel contextual information of the input image itself

to enhance the contrast. Lee et al. [4] developed a contrast enhancement method based on layer difference representation, used a tree diagram to represent multiple layers of grayscale and selected the values of grayscale difference that occur more frequently in the input, and an aggregate transformation function was designed to unify all layers. Wang et al. [5] presented a method to increase the contrast of an image, which changed the histogram of the image by weighting and setting a threshold and then performed histogram equalization to enhance the image. Abdullah-Al-Wadud et al. [6] presented a low-light enhancement method using dynamic histogram equalization that could enhance an image without losing details in the process. Although the histogram equalization-based approaches have made some contributions to the low-light enhancement tasks, they are limited by their theoretical limitations, and histogram equalization has failed to gain the same long-term attention as Retinex theory when dealing with complex low-light conditions.

B. The Retinex-Based Methods

The Retinex-based methods decompose the low-light image into a reflection component and an illumination component by a specific prior or regularization, and the estimated reflection component is often treated as the enhancement result. D. J. Jobson et al. [7] attempted to define the actual implementation of the Retinex by a balance between reproduction controlled by surround space constancy and dynamic range compression, without paying special attention to the validity of the model for color luminance and human perception. To achieve simultaneous color/luminance reproduction and dynamic compression, Jobson further proposed the Multi-Scale Retinex (MSR) [8] and multi-scale Retinex with color restoration (MSRCR) [9]. Wang et al. [10] proposed an enhancement algorithm for non-uniformly illuminated images that could provide better visual effects by enhancing image details while maintaining naturalness. Fu et al. [11] designed a new low-light enhancement method that estimated the reflectance and illumination of the observed image by a new weighted variational model that could better estimate reflectance and retain more details. Guo et al. [12] developed an effective low-light image enhancement method (LIME), which estimated the illumination of individual pixels by finding the maximum value in the R, G, B channels. Fu et al. [13] presented a fusion-based low-light enhancement method, which enhanced the image by decomposing the image into a reflective part and an illuminated part based on a light estimation algorithm and using two inputs obtained by sigmoid function and adaptive histogram equalization and then fused the two inputs with the corresponding weights in a multiscale manner. Retinex-based methods strongly rely on reliable priors and regularization to decompose low-light images. However, the Retinex-based methods often require a reliable prior and accurate estimate of the reflection component, and it is very difficult to obtain a reliable prior and accurate estimation in complex low-light conditions.

C. The Learning-Based Methods

Low-light enhancement methods based on deep learning have received the most attention in recent years, showing significant improvements in accuracy, global robustness, and visual effects.

Lore et al. [14] proposed a method to enhance low-light images using a deep autoencoder that identifies signal features from low-light images and adaptively accomplishes image enhancement and avoids over-amplifying the brighter parts of the high dynamic range images during the enhancement process. Wei et al. [15] developed a Retinex-Net based on deep learning for low-light enhancement, which includes Enhance-Net and Decom-Net, the former for decomposing low-light images and the latter for achieving image enhancement, the first real environment benchmark low-light dataset LOL was proposed. Cai et al. [16] proposed a deep learning-based end-to-end network and constructed a large multi-exposure dataset SICE, which contains 589 sets of multi-exposure sequences and the corresponding 4413 high-resolution images. Wang et al. [17] designed a global illumination aware and detail-preserving network (GLADNet) that used global illumination estimates computed on low-light images as a guide to adjust the illumination and used the original images to provide supplementary details. Zhang et al. [18] developed a low-light enhancement method based on Retinex theory kindling the darkness (KIND), which was divided into two parts, illumination, and reflection, with the former responsible for adjusting luminance and the latter for complex removal of degradation. Lv et al. [19] designed a multi-branch low-light enhancement network (MBLEN), which could extract features at different levels and achieve primary enhancement via multiple subnets, and finally complete image reconstruction by multi-branch fusion. Guo et al. [20] presented a new method called zero-reference deep curve estimation (Zero-DCE) for estimating curves, which was designed to adjust the pixel and higher-order curves of images guided by a non-reference loss function taking into account the pixel range, monotonicity, and other factors. Moran et al. [21] proposed a deep neural network named deep local parametric filters (DeepLPF), which regressed the parameters of three spatially localized filters, elliptic, fading, and polynomial filters, and adaptively applies different filters to achieve image enhancement. Liu et al. [22] designed the Retinex-inspired unrolling with architecture search (RUAS), which first characterized the internal underexposure structure of low-light images based on the Retinex model, and then optimized the process and finally completed the construction of the overall propagation structure. Lu et al. [23] developed a two-branch exposure fusion network, which consists of two parts, one for enhancement and fusion of images with different distortion levels through two branches, and one for denoising through two stages. Zhang et al. [24] proposed a method that could be implemented to enhance the temporal continuity of low-light video from static low-light images, which could judge the optical flow from a single image and generate a video sequence. Li et al. [25] introduced a luminance-aware pyramid network (LPNet) based on a coarse-to-fine strategy, which included a luminance-aware branch and two feature extraction branches for low-light image enhancement. Li et al. [26] proposed a progressive recursive image enhancement network (PRIEN), which used repeated iterations of recursive units to extract image features. The recurrent unit consists of two parts, recursive and residual blocks, where the global features of degraded images is firstly acquired using a double attention

module, and then shared deep features are passed through recurrent layers across stages to achieve image enhancement. Zamir et al. [27] developed a deep-learning low-light enhancement method that achieved low-light image enhancement by reflecting the learned properties of well-exposed images in low-light environment images. Lamba et al. [28] suggested a deep learning low-light field restoration network (L3Fnet) for low-light field (L3F) enhancement, which consists of two phases, encoding the geometric structure of all light fields in the first stage and using the acquired encoded information for image reconstruction in the second stage. Guo et al. [29] presented an end-to-end network combining target detection and low-light enhancement, which selected different enhancement subnetworks for different samples, and the network consists of two phases: enhancement and detection. Liu et al. [30] proposed an end-to-end network for converting low-light image enhancement into local affine color transformations and a noise processing approach for learning noise-decoupled affine models. Jiang et al. [31] came up with a self-regularization method based on Retinex theory, which applies Retinex to luminance while preserving image hue and saturation. Yang et al. [32] designed an end-to-end signal prior-guided layer separation and data-driven mapping network with layer-specified constraints based on Retinex theory. Karadeniz et al. [33] presented a framework for low-light enhancement that was designed progressively from coarse to fine, where first a coarse network was used to predict the low-resolution and denoised original image, and then a fine recovery network was used to generate the enhanced image. Ren et al. [73] proposed a hybrid enhancement algorithm that estimates the global information of low-light images through content streams and models edge details through spatially variant recurrent networks. Luo et al. [74] designed a curve estimation network cascaded with shared weights on the hue (H) and saturation (S) channels for an under-display camera. Though deep learning methods based on convolutional neural networks have made great progress in low-light enhancement tasks, most current methods still enhance images with inaccurate exposures, unrealistic colors, and other issues such as artifacts and noise.

D. Other Methods

Dong et al. [34] designed a new low-light enhancement method that first inverted the input low-light video and then used an image defogging algorithm to achieve the enhancement. Ying et al. [35] proposed a multi-exposure fusion framework for low-light enhancement, where the network first designed a weight matrix for image fusion by light estimation, then synthesized images with different exposures using the corresponding model of the camera, and finally used the weight matrix to fuse with the synthesized image and the input image to complete image enhancement.

E. Vision Transformer

Transformer was first proposed in the field of natural language processing (NLP) [36], with the increasing maturity of transformer in natural language processing tasks, transformer-based methods have achieved dominance in various NLP tasks [37].

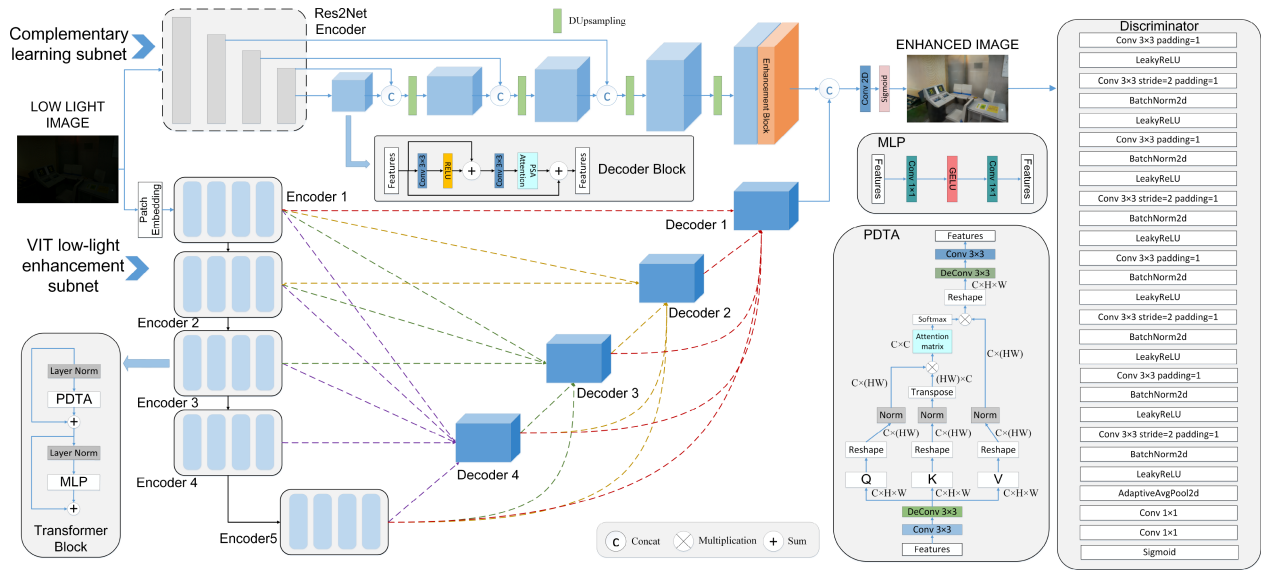


Fig. 1. Architecture of the proposed LightingNet.

Encouraged by the great success of Transformers in NLP, researchers have turned their attention to vision tasks. The image recognition task was the first application of vision transformer (ViT) [38], where ViT showed excellent speed and accuracy. Compared to traditional convolutional neural networks, ViT has shown great potential in a variety of vision tasks. Liu et al. [39] designed a hierarchical transformer (Swin Transformer) with shifted windows to limit the computation of self-attention, which serves as a general-purpose backbone for various vision tasks. Zamir et al. [40] proposed a novel transformer that could reduce the computational burden while capturing long-range pixel interactions, showing outstanding performance in several image restoration tasks. However, how to utilize both the benefits of the vision transformer and CNN for low-light enhancement remains a special challenge.

F. The Proposed Method

In contrast to other low-light enhancement methods that rely on a variety of physical priors: histogram equilibrium, Retinex model, and camera response model, the proposed LightingNet does not rely on any physical priors, but directly learns a mapping from low-light images to normally exposed images. To obtain a reliable mapping, LightingNet adopts the concept of integrated learning as follows: a CNN-based complementary learning sub-network is combined with a visual transformer-based enhancement subnetwork, where the former provides global features and the latter provides fine-grained features of the current data.

III. METHODOLOGY

In this section, we introduce the proposed LightingNet for low-light enhancement. We first detail the network architecture, including the learning complementary subnetwork and the vision transformer low-light enhancement subnetwork, and then illustrate the loss function and experimental settings. The architecture of the proposed LightingNet is shown in Fig. 1.

A. General Network Architecture

The proposed LightingNet is designed to learn a high-quality mapping from low-light image to ground truth directly, an integrated learning strategy is adopted for 3 reasons: 1) Each network has its own limitations. 2) A strong network in fitting the training data may lose its robustness in complex environments. 3) An individual complementary network, even if loaded with pre-trained weights loaded in other large dataset tasks, would hardly provide local features that satisfy the current data evaluation. LightingNet consists of two core components: 1) a VIT low-light enhancement subnet and 2) a complementary learning subnet. The VIT low-light enhancement subnet focuses on learning and fitting the current training data to provide high-level local feature information. Inspired by TDN [41] which used traditional convolutional neural networks combined with transfer learning for image dehazing, the complementary learning subnet focused on global feature extraction of low-light images using weights pre-trained in ImageNet to obtain complementary information for the VIT enhancement subnet through a specially designed decoder section. The global and local feature information acquired by two sub-networks is fused at the end of the network to achieve the multi-context fusion, leading to high-quality image enhancement.

B. Vision Transformer Low-Light Enhancement Subnetwork

The vision transformer low-light enhancement subnetwork is designed with the objective of learning the distribution and complex patterns of the local training data. We introduce the visual transformer to fit the current training data and design a full-scale structure as Unet3+ for four main reasons: 1) Unlike the convolutional layer with a limited receptive field, the visual transformer subnet can effectively capture the long-range features of the training data. 2) The presence of multiple sets of downsampling and upsampling operations in the transformer subnet structure allows the vision transformer subnet to have

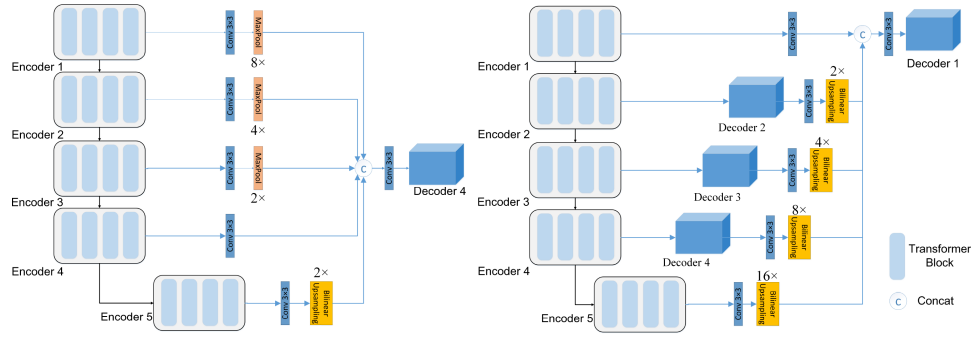


Fig. 2. Construction of the decoder 1 and decoder 4.

encoder-decoder layers of different scales/resolutions, and each set of transformer encoders has a relatively larger receptive field when processing lower resolution feature maps, making the visual transformer subnet not only limited to fit the features of original resolution but also remains sensitive on important features at multiple resolutions. 3) In each encoder-decoder layer of the visual transformer subnet with lower resolution, the same transformer block can process the features at the corresponding scale with lower resource consumption. 4) A visual transformer subnet designed based on the full-scale structure can obtain feature information of all other scales at each decoder layer, the fine features from multiple scales are preserved as much as possible in the decoder up-sampling path. In the semantic segmentation task, feature maps at different scales can enhance very different information for the network [42], [43], and these acquired fine features are often gradually lost due to upsampling and downsampling, and both the traditional Unet and Unet++ structures cannot fully utilize all the feature information at different scales [44], inspired by UNET3+ [44] and Restormer [40], we designed the VIT low-light enhancement subnet, where full skip-connection and internal connection are applied to integrate low-level contextual details and high-level features to ensure that the effective information at each decoding layer can be fully utilized at different scales.

The VIT low-light enhancement subnet consists of five transformer-based encoders at each scale, each encoder consists of four transformer blocks, each transformer block contains PDTA and MLP and two norm layers, and the low-light images first pass through the patch embedding layer to expand the channels to 24 (initial channel setting), and then enters the level 1 encoder, the scale of level 1 decoder feature map is consistent with the input image, these feature maps are then expanded and downsampled into the level 2 encoder by 3×3 convolution with stride 2. A total of four down-sampling operations are performed in the encoder path of the network, and the number of channels is also doubled while down-sampling is performed to maintain the network efficiency. To fully process the feature information of different channel numbers, the number of heads of each transformer block is also increased simultaneously after each down-sampling operation. Take Encoder 5 as an example, the number of channels of the feature map is 384 ($16 \times$ initial setting), which is divided into 8 groups with 48 channels (heads are set to 8) for processing in the transformer. In the decoder

path of VIT low-light enhancement subnet, each decoder is constructed by a different combination of encoders and decoders to obtain high-level context.

As shown in Fig. 2, taking level 4 decoder and level 1 decoder as examples, all inputs of level 4 decoder are composed of encoders, encoder 4 feature information of the same scale is directly accepted by decoder 4 firstly, while feature maps of higher scale level 1, level 2, and level 3 encoding layers are unified by corresponding different multiples of downsampling and 3×3 convolution to reach the same scale as the level 4 decoder, and the low-scale level 5 encoding layer is integrated with the other four groups of processed feature maps through bilinear upsampling and 3×3 convolution, generating feature maps with total channels of $5 \times$ initial channel, then 3×3 convolution with $5 \times$ initial channel is further applied to merge the fine-tuned features and multi-scale information. The difference between Unet3+ is that the BN and RELU activation layers are discarded when fusing the features, for applying BN and RELU activation layers at the decoder will reduce the redundancy while significantly reducing the fine information obtained by the VIT encoder, resulting in serious degradation of the network performance. Unlike the level 4 decoder, the input of the level 1 decoder includes feature information from only two encoders—the level 1 encoder and the level 5 encoder, and the remaining three components came from the decoding layers that are integrated into the level 1 scale. The processing from coarse to fine in the decoder path ensures the network’s effectiveness in integrating the full-scale exquisite information.

C. Transformer Architecture

The transformer block is the core of the VIT low-light enhancement subnet. Each transformer block consists of paired-deconv transposed attention (PDTA), multi-layer perceptron (MLP), and two norm layers. Conventional transposed attention (MDTA) [40] uses two 1×1 convolutions to expand the features and a 3×3 depth-wise convolution to emphasize the local context, and then generates an implied attention matrix $C \times C$ by computing the cross-channel covariance (Input features $C \times H \times W$; C: channel, H: height, W: width). As shown in Fig. 1, in this paper, we design an improved PDTA based on MDTA with two structural changes to improve the SA performance: 1) Replacing the 1×1 convolution with a 3×3 unbiased

convolution to expand and contract the features. 2) Placing additional depth-wise convolution in the tail to decode the context. Compared with 1×1 convolution, 3×3 convolution maintains a lower computational burden with a significantly larger receptive field, providing a more refined information flow for transposed self-attention to generate implicit attention maps by computing cross-channel covariance. The depth-wise convolution placed after the first convolution is used to encode the global and local context, while the other depth-wise convolution is placed after the transposed self-attention to decode the processed context, providing a more fine-grained flow. We adopt typical MLP structure [38], [47], which includes only two 1×1 convolution layers and GELU non-linearity [48]. The first convolution is used to expand the channel of the input feature map, the hidden layer expansion rate is set to 2.66, and where the GELU non-linearity is set to filter out the redundancy, the second convolution is designed to reduce the size of the expanded feature map to the input channel.

D. Complementary Learning Subnetwork

As a reinforcement branch to the VIT low-light enhancement subnetwork, the complementary learning subnetwork is introduced to provide globally robust features that are processed by a carefully designed decoder architecture. Specifically, the learning complementary subnet first extracts global robust feature information of low-light images through the encoder part of Res2Net [49] pre-trained on ImageNet. The encoder structure of Res2Net consists of four downsampling operations, and these robust global features are accepted by the decoder with an attention mechanism: each containing two 3×3 convolution layers, a RELU activation layer, and the polarized self-attention module PSA [50]. Up-sampling is performed between each decoder level by DUpsampling [51]. Due to the nature of Res2Net itself, it is difficult to construct a full-scale skip connection between the decoding layers and the encoding layers, to reduce the missing information due to the lack of level 1 encoder-decoder skip-connection, the output part of level 5 decoder and the output feature maps of Res2Net are integrated through concatenation, and the skip-connection with the encoder layer is set in both Level 4 and Level 3 decoder layers to realize the integration of low-level information of encoder and high-level information of decoder. The output features of the level 1 decoder are finally fused with VIT low-light enhancement subnet through an effective enhancement module [52] to complete image reconstruction.

E. Discriminator

The discriminator is used to discriminate whether the image generated by the proposed LightingNet is a true normal image or not, while the LightingNet is dedicated to generating enhanced images that the discriminator cannot identify as true or false. The structure of the discriminator includes eight 3×3 convolutional layers and two 1×1 convolutional layers, and the probability mapping is obtained by the sigmoid function. The detailed structure is shown in Fig. 1.

F. Loss Function

To cooperate with the VIT low-light enhancement subnetwork and the complementary learning sub-network to generate visually comfortable enhanced images with higher objective evaluation metrics, we adopt a hybrid loss function [71], which includes the smoothed L_1 loss, the adversarial loss function, the perceptual loss function, and the multi-scale structural similarity loss function. The hybrid loss function L can be defined as:

$$L = W_1 L_S + W_2 L_{PERC} + W_3 L_{ADV} + W_4 L_{MS}, \quad (1)$$

where W_1 , W_2 , W_3 and W_4 are the weights used to balance the hybrid loss function. The perceptual loss function [58] can provide supplementary supervision in improving the visibility of the images generated by the model. The perceptual loss function is defined as:

$$L_{PERC} = \frac{1}{N} \sum_i \frac{1}{C_i H_i W_i} \|\psi_i(G(Low)) - \psi_i(GT)\|_2^2, \quad (2)$$

where Low , GT , and N are the low-light input image, the supervised image with normal exposure, and the number of features being used in the perceptual loss function, respectively. $G(\cdot)$ is the generator, and $G(x)$ is the enhanced image. $\psi_i(\cdot)$ denotes the pre-trained network on the classification task [59]. The adversarial loss function can provide advanced supervision to the network beyond pixel-wise supervision, the adversarial loss function is defined as:

$$L_{ADV} = \sum_{n=1}^N -\log D(G(Low)), \quad (3)$$

where Low , $G(\cdot)$ and $G(Low)$ denote the low-light image, the generator and the enhanced image, respectively. $D(\cdot)$ is the discriminator, $D(G(Low))$ represents the possibility that the discriminator accepts the enhanced image to be the ground truth. The smooth L_1 loss function and the multi-scale structural similarity loss function push the network to generate images as close to the ground truth as possible at the pixel-level, where the smooth L_1 loss function is defined as:

$$L_S = \frac{1}{N} \sum_j smoothL_1(y_j - G(x_j)), \quad (4)$$

$$smoothL_1(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \quad (5)$$

where x_j and y_j denote the low-light image and the corresponding ground truth with normal exposure at pixel j location, respectively. $G(\cdot)$ is the generator. The multiscale structural similarity loss function [44] is defined as:

$$L_{MS} = 1 - \prod_{m=1}^M \left(\frac{2\mu_H\mu_G + C_1}{\mu_H^2 + \mu_G^2 + C_1} \right)^{\beta_m} \left(\frac{2\sigma_H\sigma_G + C_2}{\sigma_H^2 + \sigma_G^2 + C_2} \right)^{\gamma_m} \quad (6)$$

where H and G denote windows with the same size at the center pixel i , respectively. Gaussian filters are applied to H and G for calculation of μ_H , μ_G . M , β_m and γ_m represent the number of different scales, and the relative importance of two different

components. The C_1 and C_2 are constant variables which are set to avoid division by 0.

IV. EXPERIMENTS AND ANALYSIS

In this section, we present the experimental details and demonstrate the effectiveness of LightingNet for low-light enhancement tasks through a large number of experiments.

A. Datasets

To fully compare and demonstrate the performance of the proposed LightingNet for the low-light enhancement task, we use reference datasets to train the network and compare the subjective effects on multiple non-reference datasets, where the reference datasets are LOL [53] and SICE [16], LOL is the first real low-light benchmark dataset, including 500 pairs of low-light/normal exposure images, which includes 485 pairs of the official training set and 15 pairs of the official test set. SICE is a high-resolution multi-exposure dataset, consisting of 589 sets of multi-exposure images. Experimentally, 200 pairs of low-light/normal-exposure images were selected for training and tested on 30 pairs of low-light/normal-exposure images, all the test images were strictly excluded from training. The test images of SICE are uniformly resized to 1024×1024 . The Non-reference datasets include ExDARK [54], DICM [55], LIME [56], MEF [57], and NPE [10]. EXDARK contains 7363 unpaired low-light images, 20 of which are randomly selected for comparison by the experiment. DICM contains 64 unpaired images, LIME contains 10 unpaired images, MEF contains 17 unpaired images, NPE contains 84 unpaired images. The most degraded low-light images from the DICM, LIME, MEF, and NPE datasets were randomly selected by the experiment for comparison.

B. Experiment Setup

In the experiments, in order to serve the overfitting problem caused by the small data set as much as possible, we set the network crop size to 128×128 and augmented the data set with random 90^{circ} , 180^{circ} , 270^{circ} rotations, horizontal and mirror flips. The experiments are based on the Pytorch framework. The training batch size is set to 20. The optimizer is chosen to be ADAM, where the cosine annealing learning strategy is applied. The initial learning rate is set to 0.00005, the initial number of learning epoch rounds is set to 10, and the epoch extension multiplier is set to 2. We configure the LightingNet as follows: Embedding dim (initial channel setting) is set to 24, VIT low light enhancement subnet contains the number of heads from level 1 to level 5 set to [1, 2, 4, 8, 8], respectively, the number of Transformer blocks at each level of the encoder is set to 4. The hyperparameters W_1 , W_2 , W_3 and W_4 of the hybrid loss function are set to 0.5, 0.1, 0.0002, and 0.5, respectively. All experiments were done on NVIDIA RTX 3090 GPU.

C. Baseline Methods

To comprehensively evaluate the effectiveness of the proposed LightingNet both subjectively and objectively, we

compared it with current 14 STATE-OF-THE-ART low-light enhancement methods, including ZERO-DCE [20], LTC [24], TBEFN [23], RUAS [22], DEEPLPF [21], MBLLEN [19], KIND [18], BIMEF [35], LIME [12], FEA [34], MSRCR [9], NPE [10], SRIE [45], MF [13].

D. Evaluation Criterion

In order to objectively evaluate the proposed LightingNet from multiple aspects, we used multiple referenced evaluation metrics: Peak signal-noise ratio (PSNR) [60], Structural similarity (SSIM) [60], Pixel-based visual information fidelity (VIF) [61], Universal quality image index (UQI) [62], Multi-scale structural similarity index (MS-SSIM) [63], Spatial correlation coefficient (SCC) [64], Spectral angle mapper (SAM) [65].

E. Visual Comparison With State-of-the-Art on Referenced Datasets

To fully demonstrate the effectiveness of the proposed model on the low-light enhancement task, we compare the proposed LightingNet with its comparison methods in detail from global to local, including four aspects: color, noise, brightening accuracy, and image detail.

The global visual comparison of the LOL [53] dataset is shown in Fig. 3, where LOW is the input low-light image and GT represents the normally exposed Ground Truth. DeepLPF [21] fails to achieve acceptable brightening, loses dark details more seriously, and produces a certain degree of artifacts. NPE [10] and RUAS [22] improve in brightness, but the overall image is still dark, and difficult to reach an acceptable visual level. Images enhanced by MF [13], Zero-DCE [20] reached an acceptable range of brightness, but there is still a certain distance from the real image, and MF failed to control the amplification of noise. Zero-DCE performs better than MF in terms of color and detail. MSRCR [9] over-brightened the image, resulting in simultaneous amplification of image noise to an unacceptable level, and even loss of color in complex environments. TBEFN [23], LTC [24] are able to brighten the image correctly compared to the previous methods, where TBEFN is better in terms of color and detail, but LTC is better in terms of noise control. However, there are still issues of lack of contrast and saturation. The proposed LightingNet significantly outperforms the other comparison methods in terms of brightening accuracy, image color, and noise control, and the enhanced images are closer to the ground truth.

The global visual comparison of the SICE dataset [16] is shown in Fig. 4. Both DeepLPF [21] and SRIE [45] failed to brighten the image correctly. DeepLPF loses dark details. SRIE ignores the highlight details while the enhanced images are accompanied by anomalous color patches. Zero-DCE [20] and LTC [24] have significant improvement in brightness, but the overall remains underexposed. The Zero-DCE-enhanced images are greenish in the outdoor environments, while the LTC has lower overall saturation and contrast. KIND [18] and FEA [34] performed well in terms of brightness, but FEA produced a large number of non-real color blocks in the shadow parts of the image,

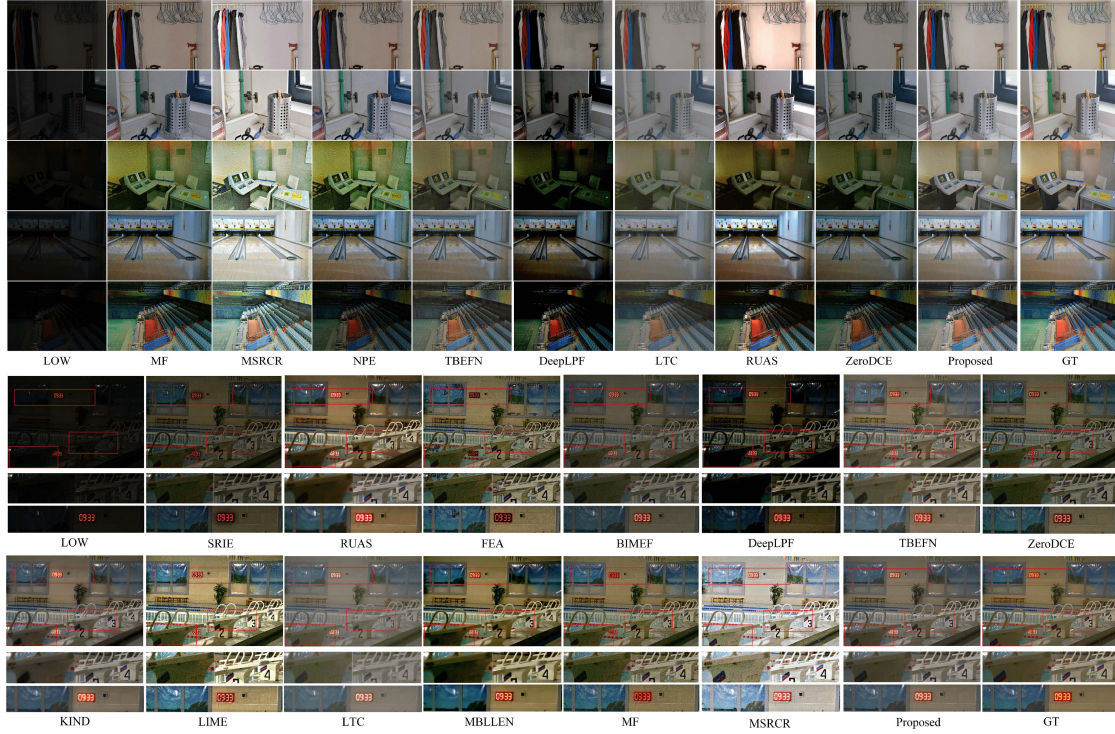


Fig. 3. Global and local visual comparison on LOL dataset.

and the color distortion is more serious. KIND's colors are closer to the real images and have better noise control, but in the well-lit parts, there were still a small number of color patches as well as artifacts. TBEFN [23] and RUAS [22] outperformed the previous methods both indoors and outdoors, but RUAS had spillover in the highlight parts and overall image contrast is slightly higher, making it less visually appealing than TBEFN, which had better color and detail and reasonable noise control. However, the shadow and highlight details of the image are also reduced, such as artifacts and railings. The proposed LightingNet significantly outperforms TBEFN in terms of color, highlight, and shadow detail, and achieves the best noise control.

The detailed comparison of the LOL dataset is shown in Fig. 3, where we select the test images from the most complex environments for comparison, and we show the details of the test images from near too far using different red boxes. As can be seen, SRIE, DeepLPP, and BIMEF failed to correctly brighten the image, DeepLPP [21] had a serious loss of dark details, and SRIE [45] had poor noise control. MBLLEN [19], MF [13], and FEA [34] have improved in image brightening, but the image noise of FEA and MF is also amplified in the process of enhancement. MBLLEN has better noise control than the previous two methods, but excessive smearing also seriously damages the image details, and all three methods fail to restore the image color accurately. LIME [12] enhanced images were all in the acceptable range in terms of brightness, but the noise control is still unsatisfactory. RUAS [22], Zero-DCE [20], LTC [24], and MSRCR [9] have different degrees of improvement in color and exposure compared to the previous methods. RUAS and LTC have better noise control than MSRCR and Zero-DCE, but RUAS images show significant oversaturation and LTC lacks

contrast. TBEFN [23] and KIND [18] not only have better color and exposure than the other comparison methods but also have acceptable noise control. LightingNet shows further improvements in noise control and image detail compared to KIND and TBEFN, almost completely removing image noise in the near shadows and maintaining more detail in the far background.

The detailed comparison of the SICE dataset [16] is shown in Fig. 4. DeePLPF [21], SRIE [45], and BIMEF [35] all fail to brighten the image correctly, and the DeepLPP-enhanced image is accompanied by artifacts and color blocks. MSR [8] and NPE [10] over-brighten the image, causing distortion of the overall visual effect. FEA [34] and KIND [18] both produced specific artifacts during the enhancement process, and the edge details of FEA also suffered from different degrees of loss. MF [13], LTC [24], and ZERO-DCE [20] enhanced images all reached an acceptable level of noise control. However, ZERO-DCE suffers from the problem of insufficient brightening, LTC suffers from low saturation, and MF loses the details of the highlights. RUAS, LIME, and TBEFN are better than the other comparison methods in color, noise control, and detail retention, but the overall image of RUAS is still too high in contrast, and LIME [12] also has the problem of highlight overflow. LightingNet outperforms other comparison methods in terms of color, highlight detail, and noise control.

F. Quantitative Comparison With State-of-The-Art on Referenced Datasets

The comparison of the quantitative evaluation metrics on the LOL and SICE datasets is shown in Table I. The proposed

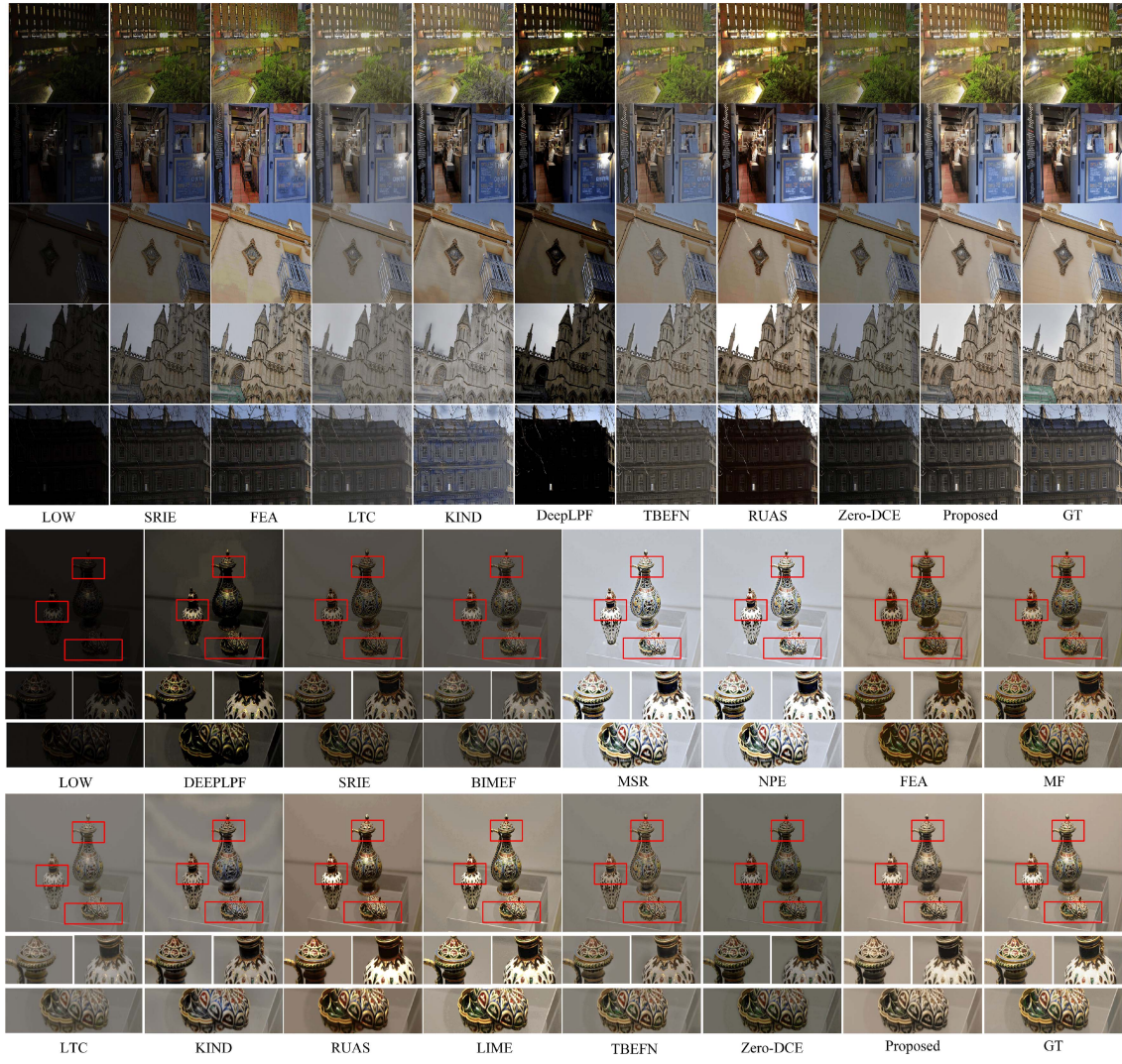


Fig. 4. Global and local visual comparison on SICE dataset.

TABLE I
QUANTITATIVE COMPARISON ON LOL [53]/SICE [16] DATASET, THE BEST RESULTS ARE MARKED AS BOLD

Method	PSNR/dB $\uparrow$	SSIM $\uparrow$	MS-SSIM $\uparrow$	UQI $\uparrow$	VIF $\uparrow$	SCC $\uparrow$	SAM $\downarrow$
DeepLPP [21]	11.691 / 10.947	0.431 / 0.396	0.786 / 0.768	0.389 / 0.336	0.300 / 0.257	0.236 / 0.214	0.407 / 0.460
SRIE [45]	11.855 / 12.013	0.495 / 0.624	0.816 / 0.834	0.498 / 0.589	0.301 / 0.305	0.288 / 0.301	0.199 / 0.188
BIMEF [35]	13.870 / 13.360	0.594 / 0.684	0.853 / 0.841	0.675 / 0.719	0.305 / 0.301	0.288 / 0.299	0.152 / 0.149
MSR [9]	13.172 / 13.960	0.461 / 0.683	0.809 / 0.841	0.761 / 0.800	0.311 / 0.329	0.282 / 0.293	0.152 / 0.282
Zero-DCE [20]	14.860 / 14.999	0.562 / 0.718	0.861 / 0.872	0.694 / 0.758	0.305 / 0.307	0.290 / 0.303	0.186 / 0.157
NPE [10]	16.969 / 17.462	0.483 / 0.731	0.836 / 0.846	0.828 / 0.873	0.306 / 0.306	0.274 / 0.284	0.208 / 0.221
LTC [24]	17.363 / 18.735	0.737 / 0.752	0.865 / 0.846	0.862 / 0.910	0.338 / 0.293	0.316 / 0.310	0.169 / 0.170
KIND [18]	20.379 / 17.861	0.824 / 0.784	0.920 / 0.873	0.901 / 0.881	0.431 / 0.326	0.378 / 0.293	0.145 / 0.204
TBEFN [23]	17.350 / 18.320	0.777 / 0.810	0.907 / 0.901	0.858 / 0.910	0.388 / 0.341	0.354 / 0.315	0.164 / 0.169
MF [13]	16.996 / 17.512	0.507 / 0.743	0.867 / 0.896	0.790 / 0.830	0.295 / 0.311	0.278 / 0.295	0.194 / 0.158
RUAS [22]	16.404 / 16.728	0.503 / 0.684	0.839 / 0.883	0.714 / 0.756	0.306 / 0.286	0.188 / 0.185	0.247 / 0.234
FEA [34]	16.716 / 17.660	0.478 / 0.726	0.826 / 0.863	0.768 / 0.846	0.273 / 0.276	0.284 / 0.292	0.218 / 0.187
MBLEN [19]	17.828 / 16.137	0.724 / 0.654	0.910 / 0.882	0.785 / 0.682	0.393 / 0.308	0.302 / 0.239	0.183 / 0.252
LIME [12]	17.021 / 19.039	0.455 / 0.734	0.794 / 0.863	0.843 / 0.902	0.349 / 0.366	0.295 / 0.302	0.226 / 0.193
Proposed	22.735 / 22.389	0.842 / 0.801	0.948 / 0.929	0.918 / 0.942	0.474 / 0.326	0.422 / 0.198	0.105 / 0.124

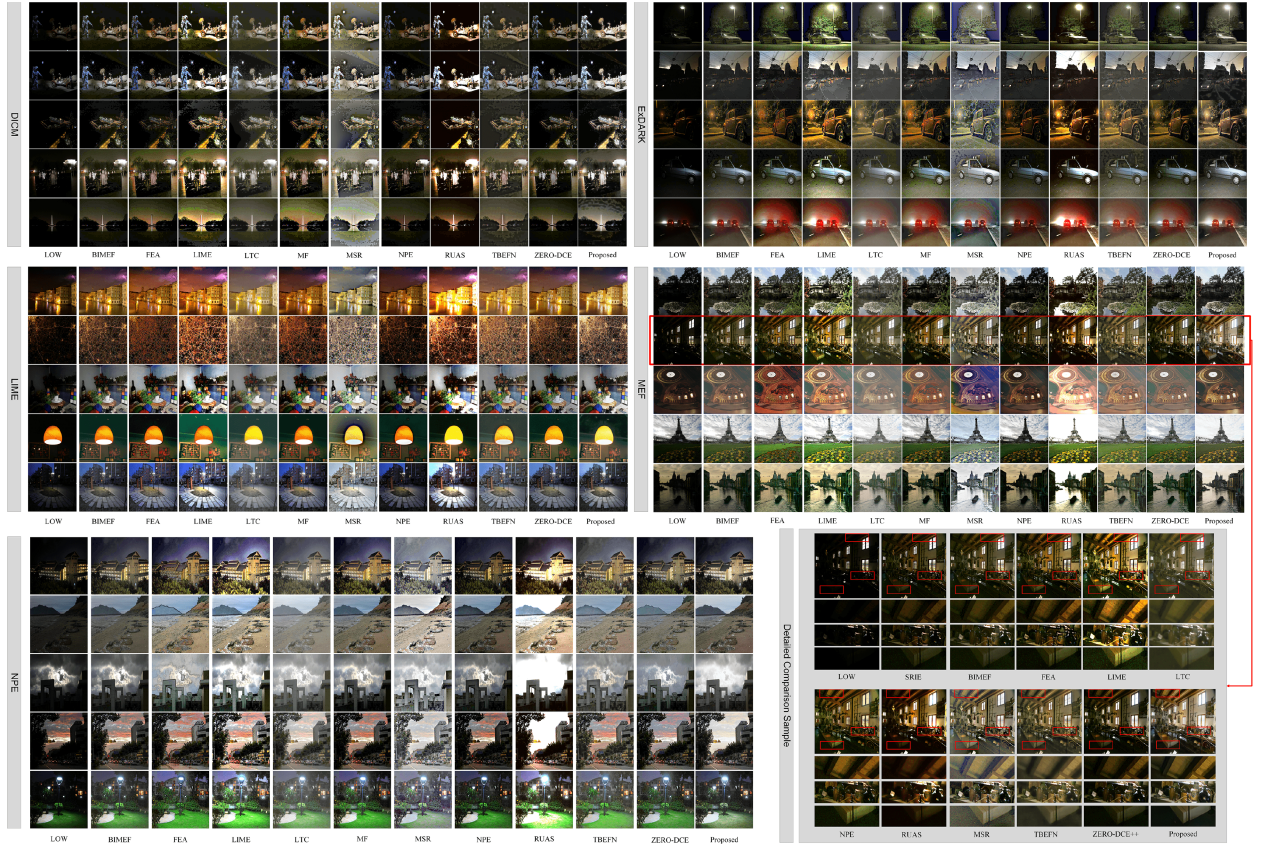


Fig. 5. Visual comparison on DICM, ExDark, Lime, MEF, and NPE dataset.

LightingNet significantly outperforms the second-place comparison method by a large margin on both datasets. In the LOL dataset, LightingNet is 2.356 dB, 0.018, 0.028, 0.017, 0.043, 0.044 and 0.040 ahead of the second-place KIND in PSNR, SSIM, MS-SSIM, UQI, VIF, SCC, and SAM, respectively. LightingNet also shows a significant advantage over the second-ranked LIME on the SICE dataset. The effectiveness of LightingNet is demonstrated in the comparison of subjective and objective evaluations.

G. Visual Comparison With State-of-the-Art on Non-Referenced Datasets

To validate the robustness of the proposed LightingNet for low-light images in different environments and illumination levels, we test it on five non-reference datasets and compare it with 8 other low-light enhancement methods, as shown in Fig. 5. LightingNet showed the best visual results in both indoor and outdoor dimly lit environments: in the LIME [56] dataset, LightingNet maximizes image noise control while preserving more dark details, and in the EXDARK [55] dataset, LightingNet-enhanced images restore dark details while minimizing image distortion compared to color blocks and artifacts produced by other comparison methods. In the church and house portions of the MEF [57] dataset, LightingNet also exhibits amazing color, pleasant visibility, and dark details compared to other

comparison methods. In the DICM [54] and NPE [10] datasets, LightingNet brightens the dark areas while retaining enough highlight detail. The effectiveness of LightingNet over other low-light enhancement methods has been demonstrated through extensive experiments.

H. Quantitative Comparison With State-of-the-Arts on Non-Referenced Datasets

We adopt four No-reference evaluation criteria to compare our model with other low-light enhancement methods, including METAIQA [67], BRISQUE [69], NIQE [70], PIQE [68], as shown in Table II, the proposed LightingNet outperformed most of the comparison methods in terms of BRISQUE and METAIQA evaluations on DCIM [54], EXDARK [55], LIME [56], and NPE [10] datasets.

I. Inference Time Discussion

In low-light image enhancement tasks, even though some algorithms can obtain reliable enhancement results, they often require a long inference time, which affects the efficiency of processing images in a limited time span. In this section, we will explore the actual performance of LightingNet and the other compared methods in terms of inference time, which is shown in Table III for the inference time results on the SICE dataset. As shown in Table III, our proposed LightingNet outperforms most

TABLE II
QUANTITATIVE COMPARISON ON EXDARK [54]/DICM [55]/LIME [56]/MEF [57]/NPE [10] DATASET, THE BEST RESULTS ARE MARKED AS BOLD

Method	EXDARK				DICM				LIME				MEF				NPE			
Metric	BRI↓	NIQ↓	PIQ↓	MET↑	BRI↓	NIQ↓	PIQ↓	MET↑	BRI↓	NIQ↓	PIQ↓	MET↑	BRI↓	NIQ↓	PIQ↓	MET↑	BRI↓	NIQ↓	PIQ↓	MET↑
SRIE	48.01	7.18	26.37	0.254	49.93	7.28	20.25	0.209	20.39	7.03	7.88	0.361	21.12	5.27	9.61	0.363	29.74	4.95	17.18	0.304
MSR	44.48	7.90	23.84	0.228	40.11	7.65	22.67	0.202	20.53	7.65	8.06	0.348	21.27	5.20	6.85	0.407	29.10	4.60	13.66	0.295
MF	53.88	8.89	21.72	0.229	52.79	8.13	20.80	0.188	20.39	7.44	6.68	0.375	18.79	5.26	6.35	0.404	29.53	5.28	13.87	0.305
FEA	55.78	8.51	20.90	0.238	53.94	9.81	18.02	0.203	24.56	7.45	8.14	0.358	25.42	5.82	6.85	0.369	39.79	6.15	9.99	0.287
LTC	46.56	6.01	33.11	0.235	54.26	5.77	22.78	0.194	30.46	7.71	15.13	0.346	30.39	5.53	21.02	0.391	32.86	4.72	24.72	0.271
BIMEF	48.89	7.15	23.58	0.265	50.48	3.81	18.02	0.203	21.11	6.81	6.83	0.402	20.80	5.09	7.82	0.412	28.75	4.90	16.91	0.317
TBEFN	33.64	6.01	22.87	0.268	32.70	5.55	17.89	0.210	21.03	7.35	9.32	0.357	21.34	5.09	12.34	0.391	24.36	4.65	15.80	0.310
RUAS	42.04	6.75	26.57	0.271	46.43	6.68	26.90	0.233	28.68	7.20	14.77	0.305	35.40	6.40	25.62	0.242	39.40	6.40	25.62	0.242
Proposed	24.56	6.35	24.83	0.304	29.20	5.79	27.31	0.232	19.23	7.46	12.49	0.381	20.14	6.03	11.26	0.413	22.94	5.12	17.98	0.344

TABLE III
INFERENCE TIME ON 1024×1024 RESOLUTION IMAGE

Method / Inference Time			
DeepLPF	1.21s	MSR	1.27s
FEA	1.01s	NPE	28.96s
KIND	0.66s	RUAS	0.56s
LIME	0.54s	SRIE	29.09s
LTC	0.26s	TBEFN	0.98s
MF	0.92s	BIMEF	0.72s
Zero-DCE	0.04s	Proposed	0.19s

TABLE IV
ABLATION ON PDTA AND LOSS FUNCTION

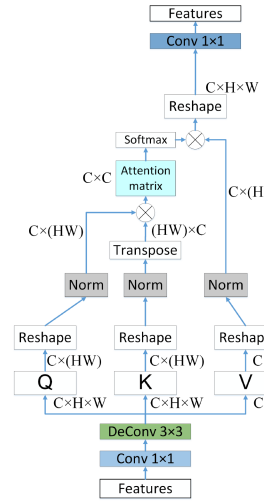
Lsmooth	Lperc	Ladv	Lms-ssim	PSNR/dB	SSIM
✓				21.740	0.818
✓	✓			22.027	0.837
✓	✓	✓		22.071	0.835
✓	✓	✓	✓	22.735	0.842
Model	Ver1	Ver2	Ver3	Ver4	
PSNR/dB	20.908	21.268	21.055	21.406	
SSIM	0.821	0.821	0.819	0.819	

comparison methods (except Zero-DCE) substantially when enhancing low-light images with the resolution of 1024×1024 .

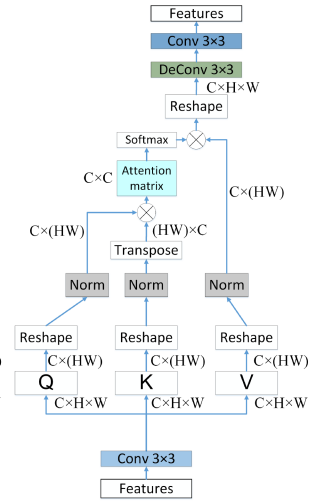
J. Ablation Study

To verify the effectiveness of the PDTA and the hybrid loss function, we conduct ablation experiments on two subjects. As shown in Table IV, the performance of the proposed LightingNet steadily increases with the addition of four loss functions, and the effectiveness of our hybrid loss function is proved. To ensure the rigor of the experiment, the complementary learning subnet is excluded from the ablation experiments to avoid possible interference. As shown in Fig. 6 and Table IV, the structure that expands and contracts the feature map by 1×1 convolution and uses depth-wise convolution to emphasize the context (Ver1) performs the least well compared to Ver3 which expands and contracts the feature map by 3×3 unbiased convolution, and we similarly ablated the placement of depth-wise convolution (Ver2, Ver3), adding depth-wise convolution to the information stream after transposed self-attention processing can better emphasize the local context in a more dense and fine information stream, while paired depth-wise convolution before and after transposed self-attention serves to encode and decode the context, further improving the network performance.

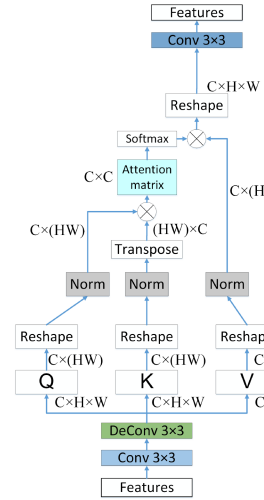
Version 1



Version 2



Version 3



Version 4

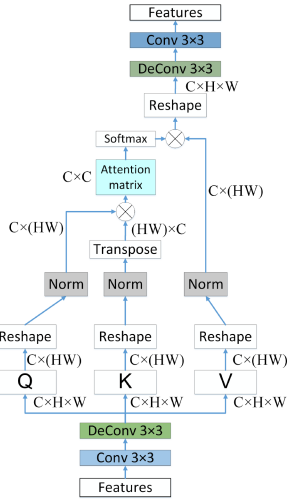


Fig. 6. Ablation study on multiple versions of PDTA.

K. Extensive Experiments

To evaluate LightingNet more comprehensively, we used YOLOv4 [66] to examine the images of LightingNet and other compared methods, and the results are shown in Fig. 7. The LightingNet-enhanced images show an overwhelming advantage over the other comparison methods, recovering the maximum amount of image details that can be identified.

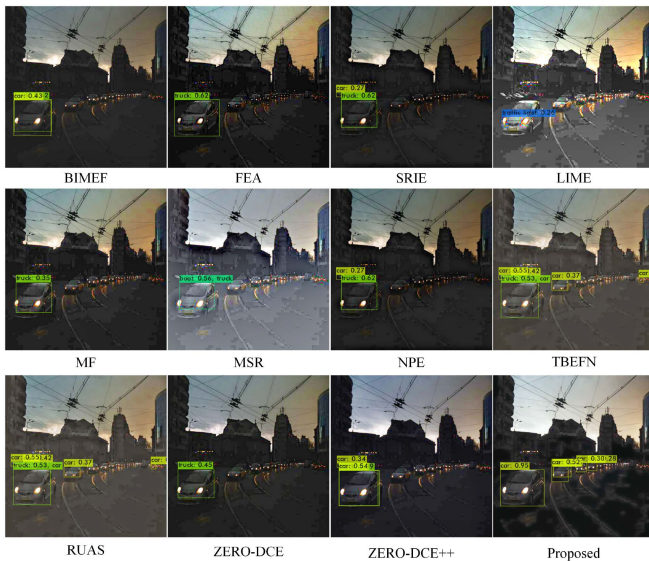


Fig. 7. Visual comparison of YOLOv4 results.

V. CONCLUSION

To address the inaccurate lightening and incomplete denoising of existing low-light enhancement methods, as well as the artifacts and color distortions generated during the enhancement process, we propose an integrated network, LightingNet, which consists of a full-scale VIT low-light enhancement sub-network and a complementary learning sub-network based on learning transfer. LightingNet outperforms the other comparison methods in various subjective and objective comparison experiments. Extensive experiments confirm the effectiveness of the proposed LightingNet.

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